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6.1200 FINAL CRIB SHEET

Contrapositive: $x \to y$, $\neg y \to \neg x$ } equivalent

Converse: $x \to y$, $y \to x$ } not equivalent.

Proof by Contrapositive: To prove: $(P \Rightarrow Q)$
$\overline{Q} \Rightarrow \overline{P}$. Assume $\overline{Q}$. Deduce $\overline{P}$.

Proof by contradiction: To prove: (P)
Assume $\overline{P}$. $(\overline{P} \Rightarrow \text{False})$

Induction:
1. Identify predicate $P(n)$.
2. Base Case(s)
3. Inductive Step:
   - > Induction: Assume $P(n)$ to prove $P(n) \Rightarrow P(n+1)$
   - > Strong Induction: Assume $P(0), P(1), \ldots, P(n)$ to prove $P(0), P(1), \ldots, P(n) \Rightarrow P(n+1)$

State Machine

Preserved predicate: True at some state + stays true from there onwards.

Invariant principle: ① Prove _ is a preserved predicate.
② We will prove that _ is an invariant.
1. True at start state. 2. True for each transition.

strictly/weakly increasing/decreasing

If f is a derived variable that
- has values in $\mathbb{N}$.
- is strictly decreasing

then the state machine terminates.
From any state s, it terminates in $\leq f(s)$ steps.

Great Common Divisor:

G1. Every common divisor of a and b divides gcd(a,b).

G2. $\gcd(ka, kb) = k \cdot \gcd(a,b)$ for all $k > 0$.

G3. If $\gcd(a,b)=1$ and $\gcd(a,c)=1$, then $\gcd(a,bc)=1$.

G4. If $a \mid bc$ and $\gcd(a,b)=1$, then $a \mid c$.

G5. If $m \mid a$ and $m \mid b$, then $m \mid \gcd(a,b)$.

G6. $\gcd(a,b) = \gcd(b, \text{rem}(a,b))$

ASYMPTOTICS

$1, \log(n), n, n\log n, n^c, c^n, n^n$

SYMBOLS	DEFINITION	INTUITION	Limit Condition
$f \in O(g)$	$\exists c. \exists n_0. \forall n \geq n_0 \mid f \mid \leq O(g)$.	"$f \leq g$" *	$\lim_{n\to\infty} \frac{f}{g} \neq \infty$
$f \in \Omega(g)$	$g \in O(f)$	"$g \leq f$" *	$\lim_{n\to\infty} \frac{f}{g} \neq 0$
$f \in \Theta(g)$	$f(n) \in O(g(n)) \cap f(n) \in \Omega(g(n))$	"$f \approx g$" *	$\lim_{n\to\infty} \frac{f}{g} =$ constant $\neq 0, \infty$
$f \in o(g)$	$\lvert f \rvert \leq o(g)$	"$f \ll g$" *	$\lim_{n\to\infty} \frac{f}{g} = 0$
$f \in \omega(g)$	$\lim_{n\to\infty} \frac{f}{g} = \infty$, aka $g(n) \in o(f(n))$	"$g \ll f$" *	$\lim_{n\to\infty} \frac{f}{g} = \infty$
$f \sim g$		"$f = g$" *	$\lim \frac{f}{g} = 1$

$(\log n)^c \ll n^c \ll c^n$

$\ln(n!) \sim n\ln(n)$

Stirling's Approximation:

$$e^{\frac{1}{(12n+1)}} \leq \frac{n!}{\sqrt{2\pi n}\left(\frac{n}{e}\right)^n} \leq e^{\frac{1}{(12n)}}$$

derivatives:

$$\frac{d}{dx}\log_a f(x) = \frac{f'(x)}{\ln(a) f(x)}$$

$$\frac{d}{dx} a^{f(x)} = a^{f(x)} \ln(a) \cdot f'(x)$$

SUMMATIONS

Perturbation Method:

$$S = 1 + x + x^2 + \ldots + x^{n-1}$$
$$xS = x + x^2 + x^3 + \ldots + x^{n-1} + x^n$$
$$S - xS = 1 - x^n$$
$$S(1-x) = 1 - x^n$$
$$S = \frac{1-x^n}{1-x}.$$

Integral Bounds:

Weakly increasing: $I + f(1) \leq S \leq I + f(n)$.

Weakly decreasing: $I + f(n) \leq S \leq I + f(1)$.

Geometric Series:

$$\sum_{k=0}^{n-1} x^k = \frac{1-x^k}{1-x}$$

not infinite: $\frac{a(1-r^n)}{1-r}$

infinite: $ar^n \to \frac{a}{1-r}$

Binet's Formula: (Fibonacci)

$$F_n = \frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right]$$

Master Theorem (Recurrences)

Plug + Chug: $k = \log_2 n$, $n = 2^k$

$$T(n) = aT\left(\left\lfloor \frac{n}{b} \right\rfloor\right) + f(n) \qquad k = \log_b a = \frac{\log a}{\log b}$$

$a \geq 1$, $b > 1$

some small constant ε

1) If $f(n) \in O(n^{k-\varepsilon}) \longrightarrow T(n) \in \Theta(n^k)$ $(k > c)$

2) If $f(n) \in \Theta(n^k) \longrightarrow T(n) \in \Theta(n^k \log n)$ $(k = c)$

3) If $f(n) \in \Omega(n^{k+\varepsilon})$ and $a \cdot f\left(\left\lfloor \frac{n}{b} \right\rfloor\right) \in O(f(n)) \to T(n) = \Theta(f(n))$ $(k < c)$.

Simple Sort: $\frac{(n-1)n}{2}$

Merge Sort: $n\log_2(n) - n + 1$
$T(n) = 2T\left(\left\lceil \frac{n}{2} \right\rceil\right) + (n-1)$

Binary Search: $T(n) = T\left(\left\lceil \frac{n}{2} \right\rceil\right) + 1$

Karatsuba: $T(n) = 3T\left(\left\lceil \frac{n}{2} \right\rceil\right) + \Theta(n)$

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$
$$\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$$
$$\sum_{k=1}^{n} k^3 = \frac{n^2(n+1)^2}{4}$$
$$\sum_{k=1}^{n} (2k-1) = n^2$$

$S \subseteq T$: set S is a subset of T

$S \subset T$: set S is a subset of T but they're not equal

Union $A \cup B$ (OR)

Intersection $A \cap B$ (AND)

Set difference $A - B$ (in A not in B)

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Divisibility: $a \mid b$ (a is a factor of b)

$\exists k \in \mathbb{Z}$ s.t. $b = ka$

$0 \mid 0$ works according to this def.

D1. If $a \mid b$, then $a \mid bc$ for all c. $\quad b \cdot c = \underbrace{(k \cdot c)}_{k'} a \quad a \mid bc$

D2. If $a \mid b$ and $b \mid c$, then $a \mid c$. $\quad \left.\begin{matrix} b = ka \\ c = ka' \end{matrix}\right\} \begin{matrix} b+c = (k+k')a \\ a \mid (b+c) \end{matrix}$

D3. If $a \mid b$ and $a \mid c$, then $a \mid sb + tc$ for all integers s and t.
$a \mid sb \qquad a \mid tc$

D4. For all $c \neq 0$, $a \mid b$ IFF $ca \mid cb$.

Modular Arithmetic: $a \equiv b \pmod n$ IFF $n \mid (a-b)$

Theorem: If $a \equiv a' \bmod n$ for any positive integer k, $a^k = (a')^k \bmod n$.

M1. $a \equiv a \pmod n$

M2. $a \equiv b \pmod n$ implies $b \equiv a \pmod n$

M3. $a \equiv b \pmod n$ and $b \equiv c \pmod n$ implies $a \equiv c \pmod n$

M4. $a \equiv b \pmod n$ implies $a + c \equiv b + c \pmod n$

M5. $a \equiv b \pmod n$ implies $ac \equiv bc \pmod n$

M6. $a \equiv b \pmod n$ and $c \equiv d \pmod n$ imply $a + c \equiv b + d \pmod n$

M7. $a \equiv b \pmod n$ and $c \equiv d \pmod n$ imply $ac \equiv bd \pmod n$

Warning: it is *not* the case that $ak \equiv bk \pmod n$ implies $a \equiv b \pmod n$ in general. It *is* true however if $\gcd(n, k) = 1$; for example, if n is prime and k is not a multiple of n.

Euclid's Algorithm: Let $a \geq b$.

lemma 1: $\gcd(a,b) = \gcd(a, b-a)$

lemma 2: $\gcd(a,b) = \gcd(a, b \text{ rem } a)$

Start state: (a,b), $a \geq b \geq 0$, not both 0.

Transition: $(x,y) \to (y, x \text{ rem } y)$

Stop when $y=0$ and output x as the gcd of a and b.

($\log(n)$ steps to termination)

Theorem (Bezout): $\gcd(a,b)$ is an integer linear combination of a & b.

Extended Euclidean Algorithm / Pulverizer:

Finds x,y such that $xa + yb = \gcd(a,b)$ $\quad a=17,\ b=3$

a	p	q	rem(p,q)	u v
	17	3	2	$2 = 1 \cdot \overset{17}{a} + -5 \cdot \overset{3}{b}$ ($17 = 3 \cdot 5 + 2$)
	3	2	$\boxed{1}$	$1 = -1 \cdot \overset{17}{a} + 6 \cdot \overset{3}{b}$
	2	1	0 gcd	

Fermat's Little Theorem:

If p is prime, $a \not\equiv 0 \pmod p$, then $a^{p-1} \equiv 1 \pmod p$.

Graphs: $V = \{a,b,c,d,e\}$

Simple graph → V is a nonempty set of vertices & E is a set of 2 element subsets of V. $\quad E = \{\{a,b\}, \{a,c\}, \{b,c\} \ldots\}$

Graph Coloring:
1. Order vertices
2. Order colors
3. For every vertex, use first legal color.

For a complete graph → max degree of K_n: $n-1$; chromatic # of K_n: n — can't do better than max degree $+1$.

Walk: length → # of edges

Path: no repeated vertices

Trail: no repeated edges — Eucerian if it uses every edge once and goes to every v.

Tour: closed trail

Cycle: tour that doesn't repeat vertices

Has Euler tour IFF connected & every v has even degree.

RSA:

Starts off → 2 large primes p, q

$n = p \cdot q$

Choose e with $\gcd(e, (p-1)(q-1)) = 1$

Compute $d \equiv e^{-1} \bmod (p-1)(q-1)$

public key → (e, n) $\quad$ private key → (d, n)

Encode: $m \mapsto \text{rem}(m^e, n) = m'$

Decode: $m' \mapsto \text{rem}((m')^d, n) = m$

Useful technique: Repeated Squaring $\text{rem}(a^b, c)$

$\text{rem}(a, c)$

$\text{rem}(a^2, c) = \text{rem}(\text{rem}(a,c), c)$ $\quad$ rem(Previous remainder squared, c)

Matching:

Maximal matching → not a subset of any other matching.

Maximum matching → if $\forall M'$, $|M| \geq |M'|$

Perfect → if it has all vertices.

Stable Matching (Gale-Shapley Algorithm): Always exists for bipartite graph.

Rogue couple: (x,y) like each other more than who they're paired with.

- Terminates in at most N^2+1 days.
- Matches everyone and produces no rogue couples.

Handshake Lemma: $|E| = \sum_{v} \deg^{(in)}(v) = \sum_{v} \deg^{(out)}(v)$ $\qquad \sum_{i=1}^{n} d(v_i) = 2 \cdot |E|$

Trees:
- Connected, acyclic graph
- n vertices & $n-1$ edges

Digraphs: (directed)
- connected → can get from u to v
- Strongly connected → u, v can reach each other

Dilworth's Thm.: $\forall t \geq 0$. Every DAG on n vertices has either chain size > 0 or antichain size $\geq \frac{n}{t}$.

DAG (Directed Acyclic Graph):
- Source: vertex with in-degree of 0. (minimal)
- Sink: vertex with out-degree of 0. (maximal)

O→O→O→O

Chain: a subset of vertices in which every pair is comparable.

Antichain: " " " " " is uncomparable.

Tasks that can be done at same time.

Depth of a vertex v is the longest path to v.

depth(v) = 0 IFF v is minimal ($u \neq v$)

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Random Variables: total function in which domain is the outcome space.

- Conditioning → $Pr[R=2 \mid M=1] = \frac{Pr[R=2 \cap M=1]}{Pr[M=1]} = 0$
- Independence → $Pr[X=x \cap Y=y] = Pr[X=x] \cdot Pr[Y=y]$
- Mutual Independence → $Pr[X_1 = x_1, \ldots, X_n = x_n] = Pr[X_1 = x_1] \ldots Pr[X_n = x_n]$

Distributions:

Probability Mass Function (PMF) → $f(x) = Pr[X=x]$ Probability that the random variable X is x.

Cumulative Distribution Function (CDF) → $F(x) = \sum_{y \leq x} Pr[X=y]$

For indicator random variables / Bernoulli: $f(0) = p$, $f(1) = 1-p$, $F(0) = p$, $F(1) = 1$

Uniform random variable on $\{1, 2, \ldots, n\}$: $\forall i \in \{1, 2, \ldots, n\}$, $f(i) = \frac{1}{n}$, $F(i) = \frac{i}{n}$ "x is uniform"

Expected Value of a random variable / average / mean:

$$Ex(R) := \sum_{w \in S} R(w) \cdot Pr[w]$$

sum of (the random variable R associated with event w × probability of event w)

Linearity of Expectation: For r.v.s R_1 & R_2, $Ex(R_1 + R_2) = Ex(R_1) + Ex(R_2)$ (no need for independence)

Theorem: Given probability space S and events $A_1, \ldots, A_n \subseteq S$, let T be a random variable that counts the # of events that occur. Then, $Ex(T) = \sum_{i=1}^{n} Pr(A_i)$.

Variance: $Var(R) = Ex((R - Ex(R))^2) = Ex(R^2) - Ex(R)^2$ Standard Deviation: $\sigma(R) = \sqrt{Var(R)}$ $\underbrace{Ex(R^2) > Ex(R)^2}$

$Ex(R^2) = Ex(R)^2$ when $Var(R) = 0$ / R is a constant

If $R_1, \ldots, R_n$ are independent r.v.s, then $Var(R_1 + \ldots + R_n) = Var(R_1) + \ldots + Var(R_n)$

Large Deviation Bounds:

Markov: R: non-negative random variable $Pr(R \geq x) \leq \frac{Ex(R)}{x}$ $Pr[R \geq c \cdot Ex(R)] \leq \frac{1}{c}$

Chebyshev: $x > 0$, assuming pairwise independence $Pr(|R - E(R)| \geq x) \leq \frac{Var(R)}{x^2}$ $Pr[|R - Ex(R)| \geq c \cdot \sigma(R)] \leq \frac{1}{c^2}$

Chernoff: Assuming mutual independence $T_1, \ldots, T_n$ (r.v.s such that $0 \leq T_i \leq 1$)

For all $c \geq 1$, $Pr[T \geq c \cdot Ex(T)] \leq e^{-\beta(c) \cdot Ex(T)}$ $\beta(c) = c \ln c - c + 1$

Relations & Counting:

Definition 1. *A* ***(binary) relation*** *$R \subseteq A \times B$ consists of*

- *a* ***domain*** *A (can be any set),*
- *a* ***codomain*** *B (can be any set),*
- *a subset $R \subseteq A \times B$ (note the abuse of notation: we equate a relation with a set, but we do need to know what the domain and codomain are.)*

In the following, let $R \subseteq A \times B$ be a relation with domain A and codomain B.

Definition 2. *R is* ***injective*** *iff every $b \in B$ has at most 1 element a satisfying $a\,R\,b$. "Every butt is shot at most once."*

Definition 3. *R is* ***surjective*** *iff every $b \in B$ has at least 1 element a satisfying $a\,R\,b$. "Every butt is shot at least once."*

Definition 4. *R is a* ***partial function*** *iff every $a \in A$ relates to at most one $b \in B$. "Every archer shoots at most once."*

Definition 5. *A relation $R \subseteq A \times B$ is* ***total*** *iff every $a \in A$ relates to at least one $b \in B$. "Every archer shoots at least once."*

Definition 6. *A* ***bijection*** *is an injective, surjective total function. "Every archer shoots exactly once, and every butt is shot exactly once."*

In the following, let $R \subset A \times A$ be a binary relation on A.

Definition 7. *R is* ***reflexive*** *iff for every $a \in A$, $a\,R\,a$.*

Theorem 5. *If A and B are finite, and R is a total injection, then $|A| \leq |B|$.*

Proof. Every $a \in A$ has at least one arrow coming out, and every B has at most one arrow coming in, so $|A| \leq \#$ arrows $\leq |B|$. □

Theorem 6. *If A and B are finite, and R is a surjective partial function, then $|A| \geq |B|$.*

Theorem 7. *If A and B are finite, and R is a bijection, then $|A| = |B|$.*

Theorem 8. *If $A_1, \ldots, A_n$ are* ***pairwise disjoint*** *finite sets, then*

$$\left| \bigcup_{i=1}^{n} A_i \right| = \sum_{i=1}^{n} |A_i|.$$

Definition 8. *R is* ***irreflexive*** *iff for every $a \in A$, $a \not R\, a$.*

Definition 9. *R is* ***symmetric*** *iff for every $a, b \in A$, $(a\,R\,b) \Rightarrow (b\,R\,a)$.*

Definition 10. *R is* ***asymmetric*** *iff for every $a, b \in A$, $(a\,R\,b) \Rightarrow (b \not R\, a)$.*

Definition 11. *R is* ***antisymmetric*** *iff for every $a, b \in A$, $[(a\,R\,b)$ AND $(b\,R\,a)] \Rightarrow (a = b)$.*

Definition 12. *R is* ***transitive*** *iff for every $a, b, c \in A$, $[(a\,R\,b)$ AND $(b\,R\,c)] \Rightarrow (a\,R\,c)$.*

Definition 13. *R is* ***antitransitive*** *iff for every $a, b, c \in A$, $[(a\,R\,b)$ AND $(b\,R\,c)] \Rightarrow (a \not R\, c)$.*

Definition 14. *A* ***preorder*** *is a reflexive and transitive relation.*

Definition 15. *A* ***strict preorder*** *is an irreflexive and transitive relation.*

Definition 16. *If R is a (strict) preorder, a and b are* ***comparable*** *iff $(a\,R\,b)$ OR $(b\,R\,a)$.*

Definition 17. *A (strict) preorder R is* ***total*** *iff every distinct pair $a \neq b \in A$ is comparable.*

Definition 18. *An* ***equivalence relation*** *is a symmetric preorder.*

Definition 19. *A* ***(strict) partial order*** *is an antisymmetric (strict) preorder.*

- A partial order is often called a ***weak partial order***.
- If a (strict) partial order is total, it is called a ***(strict) total order*** (without "partial").
- (Strict) total orders are often called ***linear*** instead of total.

Definition 20. *If R is an equivalence relation, then the* ***equivalence class*** *of $a \in A$ under R is the set $[a]_R := \{b \in A : a\,R\,b\}$.*

Definition 21. *The* ***reflexive closure (resp. symmetric closure, transitive closure)*** *of R is the minimal reflexive (resp. symmetric, transitive) relation containing R.*

Definition 22. *The* ***irreflexive kernel (resp. symmetric kernel)*** *of R is the maximal irreflexive (resp. symmetric) relation contained in R.*

Theorem 2. *The above are well-defined.*

Definition 23. *An* ***antitransitive kernel****, or* ***transitive reduction****, of R is a minimal antitransitive relation whose transitive closure is the transitive closure of R.*

Conditional Probability: $Pr[A \mid B] = \frac{P[A \cap B]}{P[B]}$

n choose k: $\binom{n}{k} = \frac{n(n-1)\ldots(n-k+1)}{k!} = \frac{n!}{k!(n-k)!}$

Bayes' Theorem: $Pr[A \mid B] = \frac{Pr[B \mid A] \cdot Pr[A]}{Pr[B]}$

PIE (Principle of Inclusion/Exclusion): $|A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$